Dynamic Bridging Tuning: Introduce an attention mechanism to the Adaptive Feature Bridging modules so they can dynamically decide which teacher to trust more based on the severity of the weather in different regions of the same image.

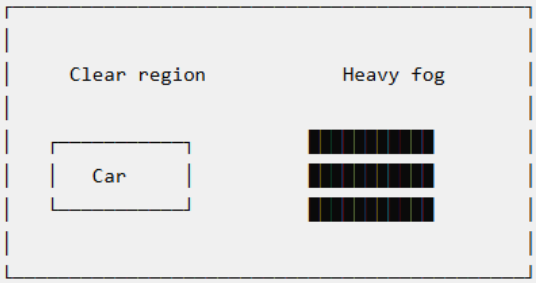
This is a **very natural extension** of DTFA, because the original framework has two teachers with different strengths, while your idea says:

> **The optimal teacher may differ not only from image to image, but even from region to region within the same image.**

Your observation is:

> Why should the bridging module treat IRT and SPT equally everywhere?

Consider one image:



In the clear region:

- SPT may provide excellent semantic information.
- IRT may not add much.

In the heavy-fog region:

- SPT's semantic representation may be unreliable.
- IRT's weather-invariant information may be more useful.

So instead of:

$$F_{student} \leftarrow F_{IRT} + F_{SPT}$$

with fixed contribution, you propose:

$$\boxed{F_{student} \leftarrow \alpha F_{IRT} + (1-\alpha) F_{SPT}}$$

where

$$\boxed{\alpha = \alpha(x, y)}$$

depends on the **spatial location**.

That is the core of your idea.

Why call it "Dynamic Bridging Tuning"

Bridging

The Adaptive Feature Bridging module already bridges:

$$\text{Teacher feature space} \leftrightarrow \text{Student feature space}$$

Tuning

You're changing how much each teacher contributes.

Dynamic

The contribution isn't fixed.

It changes according to:

$$(x, y, \text{weather condition}, \text{feature content})$$

Hence:

$$\boxed{\text{Dynamic Bridging Tuning}}$$

Where should the attention mechanism operate?

Option A — Image-level attention

Produce just one scalar:

$$\alpha \in [0, 1]$$

for the whole image.

Then:

$$F_B = \alpha F_{IRT} + (1 - \alpha) F_{SPT}$$

Example:

$$\alpha = 0.8$$

means:

> "This entire image is heavily degraded, so trust IRT more."

This is simple, but it **doesn't fully match your idea**, because weather severity can differ within the same image.

Your stronger version: spatial attention

Instead produce:

$$A \in [0,1]^{H \times W}$$

where:

$$A_{ij}$$

represents the amount of trust given to IRT at spatial location (i, j).

Then:

$$\boxed{F_B(i,j) = A_{ij} F_{IRT}(i,j) + (1 - A_{ij}) F_{SPT}(i,j)}$$

This is much closer to your proposed enhancement.

But how do we calculate this attention?

Suppose at one feature level:

$$F_{IRT}, F_{SPT}, F_S \in \mathbb{R}^{C \times H \times W}$$

First concatenate the teacher/student information:

$$F_{cat} = [F_{IRT}; F_{SPT}; F_S]$$

Therefore:

$$F_{cat} \in \mathbb{R}^{3C \times H \times W}$$

Then use a small convolutional network:

$$Z = Conv_{3\times3}(F_{cat})$$

and:

$$A = \sigma(Conv_{1\times1}(Z))$$

so:

$$A \in [0,1]^{1 \times H \times W}$$

Then dynamically fuse the teachers:

$$\boxed{F_B = A \odot F_{IRT} + (1 - A) \odot F_{SPT}}$$

where $\odot$ denotes element-wise multiplication.

Why include the student feature?

A particularly useful question is:

> Should attention depend only on the two teacher features, or also on the student?

I would strongly consider using:

$$A = g(F_{IRT}, F_{SPT}, F_S)$$

because the student itself can tell us:

> "Which teacher's information is currently more compatible with what I am trying to detect?"

For example:

- Student feature already has strong semantics → SPT may be sufficient.
- Student feature is highly corrupted → IRT may be more useful.

Thus the attention is **student-aware**.

But your phrase "based on severity of weather" deserves special treatment

Feature attention ≠ explicitly knowing weather severity.

If you simply feed features into an attention module, the network may learn some correlation with weather degradation, but it isn't explicitly measuring weather severity.

You could make your idea much more principled by introducing a **weather severity map**.

Let:

$$W(x, y) \in [0, 1]$$

represent estimated local weather severity.

For example:

$$W(x, y) = 0$$

means:

> clear region

while

$$W(x, y) = 1$$

means:

> severely degraded region.

Then you could define teacher weights based on it.

For example:

$$\alpha(x,y) = \sigma\left(\beta_0 + \beta_1 W(x,y)\right)$$

If $\beta_1 > 0$, then increasing weather severity increases IRT weight.

Therefore:

$$W \uparrow \Rightarrow \alpha \uparrow$$

and hence:

$$\text{severe weather} \Rightarrow \text{more IRT}$$

Even better: learn severity instead of explicitly labeling it

In real datasets, you usually don't have a pixel-level "fog severity" label.

So create a learned weather estimator:

$$W = f_w(I_a)$$

where:

$$f_w : \text{adverse image} \to [0,1]^{H \times W}$$

Then:

$$A = f_a(F_{IRT}, F_{SPT}, F_S, W)$$

This creates a two-stage reasoning:

$$\text{Weather degradation} \to \text{Teacher reliability}$$

This is much more interesting academically.

Another possibility: use teacher disagreement

Here's an idea that I think is particularly valuable for your paper.

Suppose the two teachers produce very different features in some region:

$$D(x,y) = \|F_{IRT}(x,y) - F_{SPT}(x,y)\|$$

Large:

$$D(x,y) \gg 0$$

means the teachers strongly disagree.

That disagreement may indicate:

- severe degradation
- ambiguous semantic information
- feature instability

You could therefore make attention depend on:

$$A = f(F_{IRT}, F_{SPT}, F_S, D)$$

This gives the attention network an explicit measure of **teacher uncertainty/disagreement**.

A very nice mathematical interpretation

Your framework can be viewed as a **mixture-of-experts** system.

We have two experts:

$$E_1 = IRT$$

$$E_2 = SPT$$

and a gating network:

$$G(x, y) = [\alpha(x, y), 1 - \alpha(x, y)]$$

Then the fused feature is:

$$F_B(x, y) = \alpha(x, y)E_1(x, y) + (1 - \alpha(x, y))E_2(x, y)$$

This is essentially:

$$\boxed{\text{Mixture of Teacher Features}}$$

where the attention/gating network decides which teacher to trust.

This theoretical connection could strengthen the motivation for your enhancement.

Scalar attention vs channel attention vs spatial attention

You have another design choice.

Suppose:

$$F \in \mathbb{R}^{C \times H \times W}$$

You can create:

Image-level attention

$$\alpha \in \mathbb{R}$$

One weight for entire image.

Spatial attention

$$A \in \mathbb{R}^{1\times H\times W}$$

One weight per location.

This matches your idea of **different regions having different weather severity**.

Channel attention

$$A \in \mathbb{R}^{C\times 1\times 1}$$

Different feature channels get different teacher weights.

For example:

$$\alpha_1 = 0.2, \quad \alpha_2 = 0.9, \quad \alpha_3 = 0.4$$

Spatial + channel attention

The strongest version is:

$$A \in \mathbb{R}^{C\times H\times W}$$

so every channel at every spatial location can have a different teacher weight.

But this is more expensive and can be harder to train.

For your particular motivation, I would start with:

$$\boxed{A \in [0,1]^{H\times W}}$$

because your argument is explicitly about **weather severity in different regions**.

The attention should probably be added AFTER feature bridging

What loss should train this attention?

Option 1 — Entropy regularization

Your attention has:

$$A(x, y)$$

and:

$$1 - A(x, y)$$

These can be interpreted as probabilities of trusting two teachers.

Define entropy:

$$H(A) = -A \log A - (1 - A) \log(1 - A)$$

This controls how uncertain the gating is.

But this alone doesn't tell the model **which teacher should win**, so it's not sufficient by itself.

Option 2 — Teacher reliability loss

Suppose we somehow estimate which teacher performs better in a region.

Define:

$$R_{IRT}(x, y)$$

and:

$$R_{SPT}(x, y)$$

as teacher reliability scores.

Then an ideal attention weight could be:

$$A^*(x, y) = \frac{R_{IRT}}{R_{IRT} + R_{SPT}}$$

and train:

$$L_{gate} = \|A - A^*\|^2$$

This is theoretically elegant, but obtaining R_{IRT} and R_{SPT} is difficult.

Option 3 — Distillation-based self-guidance

This is more practical.

Let the fused feature be:

$$F_B = A\tilde{F}_{IRT} + (1 - A)\tilde{F}_{SPT}$$

Then require the student feature F_S to be close to the fused representation:

$$L_{align} = \|F_S - F_B\|^2$$

The network itself learns which teacher combination minimizes the downstream objective.

Total objective could be:

$$\boxed{L = L_{det} + \lambda L_{align} + \gamma L_{reg}}$$

where L_{reg} prevents pathological attention behavior.

A stronger formulation: reliability-aware teacher fusion

You can formulate the entire problem as:

$$\boxed{F_B(x,y) = \frac{r_I(x,y)\tilde{F}_{IRT}(x,y) + r_S(x,y)\tilde{F}_{SPT}(x,y)}{r_I(x,y) + r_S(x,y) + \epsilon}}$$

where:

$$r_I, r_S \geq 0$$

represent learned reliability scores.

Then define:

$$\alpha = \frac{r_I}{r_I + r_S}$$

so automatically:

$$1 - \alpha = \frac{r_S}{r_I + r_S}$$

and therefore:

$$F_B = \alpha\tilde{F}_{IRT} + (1 - \alpha)\tilde{F}_{SPT}$$

This is mathematically cleaner than arbitrarily choosing two independent attention values.

Why should weather severity influence IRT weight?

This needs a theoretical justification in your paper.

Suppose the degraded observation is:

$$I_a = \mathcal{D}(I_c; w) + n$$

where:

- I_c = clear image
- $\mathcal{D}$ = weather degradation operator
- w = weather severity
- n = noise

As w increases:

$$w \uparrow \Rightarrow \text{visual degradation} \uparrow$$

and therefore the semantic detector has a more difficult job.

Because IRT is explicitly designed to capture **weather-invariant information**, its relative usefulness can increase:

$$w \uparrow \Rightarrow r_{IRT} \uparrow$$

while potentially:

$$r_{SPT} \downarrow$$

Thus:

$$w \uparrow \Rightarrow \alpha \uparrow$$

This gives you the theoretical motivation for Dynamic Bridging Tuning.